

SAARBRUCKEN, Germany

WMO: 107080

Lat: 49.2128N

Lon: 7.1078E

Elev: 1051

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	21.1	8.9	9.0	20.9	14.3	11.8	23.8	29.6	46.8	26.4	45.7	6.9	70	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	87.5	67.3	82.8	65.8	80.2	64.9	69.0	82.4	67.4	80.1	65.9	77.1	7.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	94.6	73.3	62.9	89.1	71.8	61.2	84.0	70.2	33.6	83.6	32.4	80.0	31.2	77.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	19.2	17.0	11.7	92.7	5.3	2.7	7.9	94.6	4.8	96.2	1.8	97.7	-2.1	99.6		
(5)			11.0	72.6	5.4	2.0	7.1	74.0	3.9	75.2	0.9	76.4	-3.1	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	34.8	35.8	42.8	51.1	56.5	63.6	67.0	64.9	59.1	51.7	42.8	37.2	(6)
(7)		DBStd	13.17	7.45	7.51	7.34	7.15	6.84	6.55	6.39	6.11	5.64	6.91	6.85	7.28	(7)
(8)		HDD50	1893	472	398	238	71	17	1	0	0	2	64	233	397	(8)
(9)		HDD65	5504	936	818	688	417	277	105	54	77	192	413	667	860	(9)
(10)		CDD50	2147	1	1	15	106	218	407	526	463	275	118	16	1	(10)
(11)		CDD65	284	0	0	0	1	13	62	116	75	16	1	0	0	(11)
(12)		CDH74	2498	0	0	0	24	128	537	1042	634	129	4	0	0	(12)
(13)		CDH80	738	0	0	0	1	25	145	350	199	18	0	0	0	(13)
(14)	Wind	WSAvg	7.8	9.3	8.9	9.1	7.5	7.4	6.8	6.9	6.5	6.6	7.4	8.2	9.3	(14)
(15)	Precipitation	PrecAvg	33.5	2.8	2.4	2.5	2.2	2.9	2.8	3.1	3.0	2.6	2.6	3.6	3.4	(15)
(16)		PrecMax	46.9	6.9	6.5	6.7	5.0	7.4	6.0	7.8	10.2	6.5	7.7	7.8	11.3	(16)
(17)		PrecMin	21.1	0.4	0.4	0.3	0.6	0.5	0.0	0.6	0.6	0.8	0.2	0.6	0.1	(17)
(18)		PrecStd	6.4	1.4	1.6	1.3	1.1	1.5	1.5	1.7	1.8	1.5	1.6	1.7	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	57.3	66.4	77.1	83.9	89.2	93.0	91.3	82.7	73.6	62.3	55.1	(19)
(20)			MCWB	50.5	46.7	50.9	58.9	65.8	68.8	68.7	68.4	65.1	61.1	56.1	50.0	(20)
(21)		2%	DB	50.4	51.5	62.4	73.1	77.2	84.3	87.9	85.8	78.4	69.4	57.5	51.9	(21)
(22)			MCWB	47.5	45.1	49.3	55.8	61.8	66.2	67.0	67.9	63.9	59.7	52.8	48.7	(22)
(23)		5%	DB	48.1	48.3	57.4	68.3	73.3	80.3	84.0	80.7	73.6	64.5	55.3	49.8	(23)
(24)			MCWB	45.9	44.4	47.9	54.0	59.4	65.6	66.0	65.6	61.8	57.9	51.5	46.9	(24)
(25)		10%	DB	44.9	46.0	53.7	64.2	69.4	76.6	80.2	76.8	69.7	61.1	53.2	46.8	(25)
(26)			MCWB	43.5	43.2	46.4	52.4	58.3	63.9	64.6	63.4	60.6	56.1	49.9	45.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	48.9	53.5	60.2	67.5	71.0	70.5	70.7	67.2	62.5	57.2	51.9	(27)
(28)			MCD B	53.1	52.4	61.2	73.7	80.5	85.1	84.3	85.2	78.3	69.7	60.1	53.6	(28)
(29)		2%	WB	48.8	46.5	51.0	57.3	64.0	68.4	68.4	68.4	65.0	60.4	53.9	49.3	(29)
(30)			MCD B	50.5	50.1	58.6	70.2	75.0	80.0	83.2	81.2	75.2	66.8	57.2	51.6	(30)
(31)		5%	WB	45.7	44.6	49.3	54.9	61.1	66.6	67.2	66.8	63.2	58.4	52.0	47.3	(31)
(32)			MCD B	47.3	47.7	55.7	66.2	69.5	77.2	80.9	78.4	72.2	63.7	55.2	49.2	(32)
(33)		10%	WB	43.2	42.5	47.5	53.1	58.7	64.7	65.6	65.1	61.2	56.9	49.9	45.1	(33)
(34)			MCD B	44.6	45.0	53.1	62.7	66.8	74.6	77.4	75.1	68.0	61.4	53.0	46.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	9.9	14.4	17.5	17.0	17.8	17.9	17.5	16.6	13.3	9.0	7.7	(35)
(36)			MCDBR	9.5	14.4	21.3	23.8	23.0	23.4	24.3	24.2	22.7	17.9	12.0	9.7	(36)
(37)			MCWBR	8.1	9.9	11.8	11.7	11.2	10.9	9.3	10.2	10.9	10.5	8.5	7.7	(37)
(38)		5% WB	MCD BR	9.1	12.5	17.6	21.6	20.4	20.8	21.0	20.8	19.2	15.7	11.1	9.0	(38)
(39)			MCWBR	8.2	9.2	10.6	11.1	10.6	10.4	9.0	9.6	10.0	9.8	8.7	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.334	0.361	0.389	0.399	0.403	0.400	0.396	0.371	0.355	0.328	0.306	(40)	
(41)		taud	2.436	2.391	2.327	2.280	2.279	2.299	2.296	2.334	2.380	2.420	2.440	2.429	(41)	
(42)		Ebn at Noon	238	257	268	272	273	272	271	266	262	245	226	221	(42)	
(43)		Edh at Noon	20	27	33	39	40	40	39	36	31	26	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	268	502	885	1360	1609	1811	1709	1468	1109	644	303	212	(44)	
(45)		RadStd	40	92	136	177	172	156	173	152	126	57	32	33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page